Chapter 5

Transformation Theory and Effective Media

From Jacobians to Binary Layers: Engineering Anisotropic Conductivity

Thermal Photonics: A Unified Framework for Engineering Heat Flux
Book 3 of the Open-Source Engineering Optics Trilogy

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Learning Objectives

After completing this chapter, the reader will be able to:

- ***LO-05.1*** *State the form-invariance of the steady-state heat equation under coordinate transformations and derive the transformed conductivity tensor* $\boldsymbol{\kappa}' = \mathbf{J} \cdot \boldsymbol{\kappa} \cdot \mathbf{J}^T / \det(\mathbf{J})$.
- ***LO-05.2*** *Design the five canonical transformation devices (cloak, concentrator, rotator, inverter, camouflage) by specifying the appropriate coordinate mapping and resulting* $\boldsymbol{\kappa}'$.
- ***LO-05.3*** *Quantify the discretization error when approximating a continuous conductivity profile with* N *homogeneous layers and apply DR-05.1 to select the minimum layer count for a target cloaking efficiency.*
- ***LO-05.4*** *Explain why the same Jacobian simultaneously transforms* $\boldsymbol{\kappa}$ *and* ε *(permittivity), and state the co-design constraint DR-05.2.*
- ***LO-05.5*** *Apply the Wiener bounds (parallel and series averaging) to design binary layered composites with prescribed anisotropy* $\kappa_\parallel/\kappa_\perp$.
- ***LO-05.6*** *Select IR-compatible material pairs from the design landscape and apply DR-06.1 through DR-06.3 for thermal-optical metamaterial fabrication.*
- ***LO-05.7*** *Perform a fabrication sensitivity analysis using the sensitivity coefficients* $\partial(\kappa_\parallel/\kappa_\perp)/\partial f$ *and determine whether PVD or ALD tolerances are adequate for a given design.*
- ***LO-05.8*** *Place transformation devices in the Mode-Path Matrix network context (Ch. 4) and identify when transformation thermotics is the appropriate tool for a given thermal path.*
- ***LO-05.9*** *Evaluate coating mechanical compatibility on metamaterial surfaces using the stress budget, critical cracking thickness, and CTE mismatch criteria (DR-05.7).*
- ***LO-05.10*** *Assess whether interface thermal resistance materially degrades the effective anisotropy of a binary-layer stack and apply DR-05.8 to decide when an interface correction is required.*

Abbreviations and Nomenclature

Abbreviation	**Meaning**
ALD	Atomic Layer Deposition
AR	Anti-Reflection (coating)
BCs	Boundary Conditions
CTE	Coefficient of Thermal Expansion
CVD	Chemical Vapor Deposition
DOE	Design of Experiments
DR	Design Rule
EMT	Effective Medium Theory
FDM	Finite Difference Method
FEA	Finite Element Analysis
IR	Infrared
LWIR	Long-Wave Infrared (8–14 μm)
MPM	Mode-Path Matrix
MWIR	Mid-Wave Infrared (3–5 μm)
PGM	Precision Glass Molding
PVD	Physical Vapor Deposition
WE	Worked Example

Symbol	**Meaning**	**SI Unit**
$\boldsymbol{\kappa}$	Thermal conductivity tensor	$\mathrm{W\,m^{-1}\,K^{-1}}$
$\kappa_{\parallel}$	Parallel (in-layer) conductivity	$\mathrm{W\,m^{-1}\,K^{-1}}$
$\kappa_{\perp}$	Perpendicular (cross-layer) conductivity	$\mathrm{W\,m^{-1}\,K^{-1}}$
$\kappa_{\perp,\mathrm{eff}}$	Perpendicular conductivity incl. interfaces	$\mathrm{W\,m^{-1}\,K^{-1}}$
κ_0	Background conductivity	$\mathrm{W\,m^{-1}\,K^{-1}}$
$\mathbf{J}$	Jacobian matrix of coordinate transformation	—
f	Volume fraction of material A in binary composite	—
f_{opt}	Optimal volume fraction for target anisotropy	—
N	Number of discrete layers	—
R_1	Inner radius (cloak/concentrator)	m
R_2	Outer radius (cloak/concentrator)	m
R_{cloak}	Cloaking efficiency	—
R_{conc}	Concentration ratio	—
Γ	Mode dominance ratio ($G_{\mathrm{rad}}/G_{\mathrm{cond}}$)	—
ε	Permittivity (optical) or emissivity	—
α_{CTE}	Coefficient of thermal expansion	$\mathrm{K^{-1}}$
$\alpha_{\parallel}$, $\alpha_{\perp}$	CTE parallel/perpendicular to layers	$\mathrm{K^{-1}}$
E	Young's modulus	GPa
$E_{\parallel}$, $E_{\perp}$	Modulus parallel/perpendicular to layers	GPa
σ_{res}	Residual film stress	MPa
σ_{crit}	Critical stress for cracking	MPa
t_{crit}	Critical cracking thickness	μm
ν	Poisson's ratio	—
G_{cond}	Conduction conductance	$\mathrm{W\,K^{-1}}$
G_{rad}	Radiation conductance	$\mathrm{W\,K^{-1}}$
R_{int}	Interface thermal resistance	$\mathrm{m^2\,K/W}$
S_f	Sensitivity coefficient (anisotropy to volume fraction)	—

Subscript and Superscript Conventions

Notation	Convention
$\parallel$, $\perp$	Parallel and perpendicular to layer interfaces
A, B	Constituent materials in binary composite
cond, rad	Conduction and radiation modes
eff	Effective (homogenized) value
cloak, conc, rot	Cloak, concentrator, rotator device type
res	Residual (stress)
crit	Critical (cracking threshold)
int	Interface (thermal resistance)

Numbering conventions: Design rules are DR-05.n (transformation and fabrication) and DR-06.n (material selection). Worked examples are WE-05.n. Learning objectives are LO-05.n.

Chapter Roadmap and Deliverables

Engineering Deliverable: A complete design methodology for realizing prescribed anisotropic conductivity profiles using binary layered composites, validated against fabrication tolerances, IR transparency requirements, and coating mechanical compatibility.

Figure 05.1 shows the chapter workflow spine, progressing from the mathematical foundation (transformation thermotics) through discretization, effective medium theory, material selection, and fabrication tolerance analysis. The v3 revision extends the fabrication section to include mechanical compatibility of functional coatings on metamaterial surfaces. The v4 revision adds interface resistance integration, a verification recipe, and a system-level worked example addressing all three reviewer priorities.

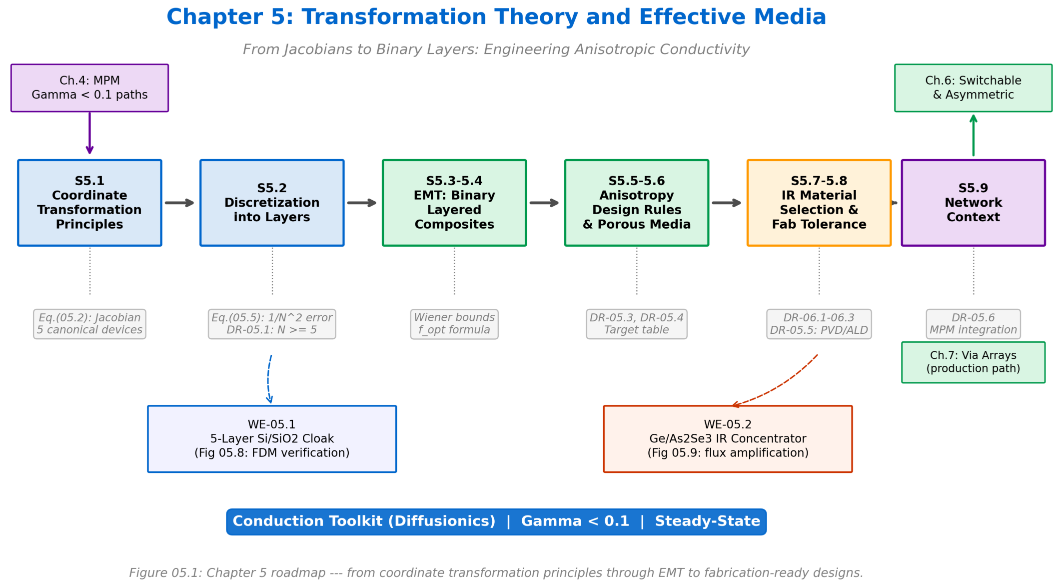

Figure 05.1: Chapter 5 roadmap: from coordinate transformation principles through discretization and EMT to fabrication-ready binary-layer designs. Each section feeds forward; the network context (§05.9) connects back to the Mode-Path Matrix of Chapter 4.

Pain Point

From tensor prescription to physical realization: Coordinate transformation yields a spatially varying conductivity tensor that enforces exterior field invariance. In its ideal form, this tensor is continuous and may approach singular limits (e.g., $\kappa_r \to 0$ at a cloaking boundary).
The engineering challenge is therefore not the mapping itself, but its realization: how to approximate a continuous tensor field using a finite, layered material architecture with bounded properties, finite thickness, and manufacturable interfaces.
This chapter develops a reproducible pipeline linking Jacobian tensor → discretization → effective-medium synthesis → material selection → tolerance and multiphysics verification.

05.1 Coordinate Transformation Principles

Why transformation theory? In practical thermal engineering, the objective is often not merely to adjust geometry or material values, but to enforce a *field-level constraint*: preserve the exterior temperature distribution while reshaping heat-flow paths internally. Typical goals include cloaking a hot region, concentrating flux into a sensing domain, or redirecting heat around a mechanically fragile component.

Classical optimization can tune shapes and isotropic conductivities, yet it does not directly guarantee invariance of the surrounding field. Transformation thermotics offers a constructive alternative: design a coordinate mapping that enforces the desired field behavior, and derive the required effective conductivity tensor through the Jacobian of that mapping.

The central task is therefore twofold: (1) demonstrate the form-invariance of the governing equation under coordinate transformation, and (2) translate the resulting tensor prescription into a fabricable material architecture—a problem addressed in the subsequent sections.

The steady-state heat equation with no internal sources is:

$$\nabla \cdot \big(\boldsymbol{\kappa} \cdot \nabla T\big) = 0 \tag{05.1}$$

This equation is *form-invariant* under smooth coordinate transformations $\mathbf{x} \to \mathbf{x}'$. Under such a mapping, the conductivity tensor transforms as:

$$\boldsymbol{\kappa}' = \frac{\mathbf{J} \cdot \boldsymbol{\kappa} \cdot \mathbf{J}^T}{\det(\mathbf{J})} \tag{05.2}$$

where $\mathbf{J} = \partial \mathbf{x}'/\partial \mathbf{x}$ is the Jacobian of the transformation. This is the master equation of transformation thermotics—every device in this chapter derives from a specific choice of coordinate mapping.

Conduction Mode (Diffusionics)

Form-invariance and Fourier's law. Equation (05.2) means that Fourier's law $\mathbf{q} = -\boldsymbol{\kappa} \cdot \nabla T$ retains its form in the transformed space, but the conductivity tensor acquires the geometric information of the mapping. This is purely a conduction-mode ($\Gamma < 0.1$) result; radiative transport does not enjoy the same simple form-invariance (but see §05.3 for the optical parallel).

05.1.1 Cloak Mapping

The thermal cloak maps the annular region $R_1 \le r' \le R_2$ from the compressed domain $0 \le r \le R_2$:

$$r' = R_1 + r\,\frac{R_2 - R_1}{R_2}, \qquad \theta' = \theta \tag{05.3}$$

The resulting conductivity in the cloak shell is:

$$\kappa_r = \kappa_0\,\frac{r' - R_1}{r'}, \qquad \kappa_\theta = \kappa_0\,\frac{r'}{r' - R_1} \tag{05.4}$$

Figure 05.2 shows the field restoration effect of a properly designed cloak: isotherms that are distorted by a bare inclusion are restored to near-parallel outside R_2.

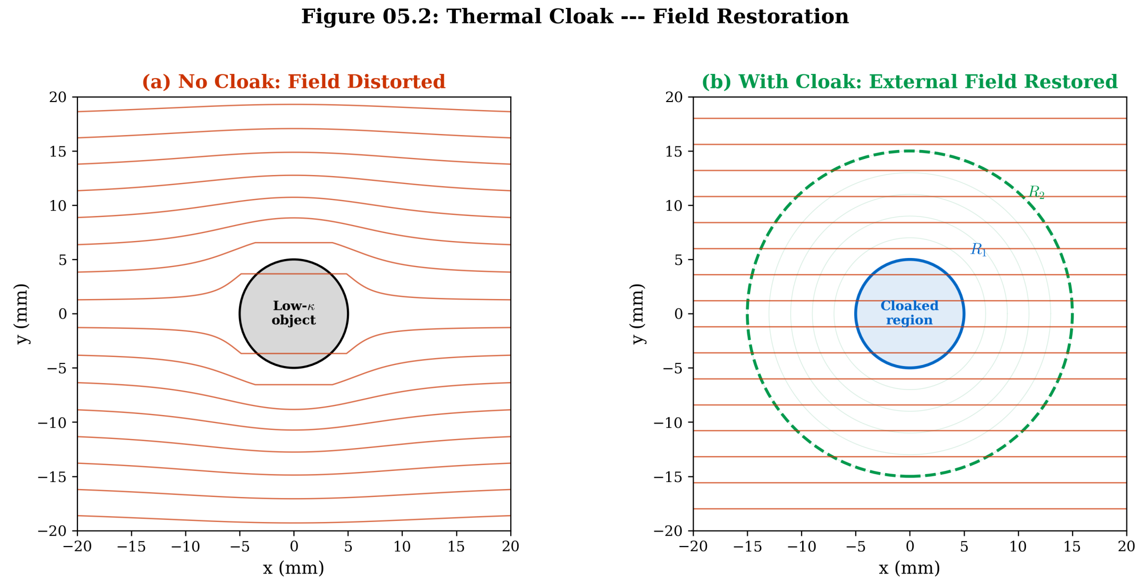

Figure 05.2: Thermal cloak field restoration (cross-checks Eqs. (05.3)–(05.4)). **(a)** Anisotropy profile κ_θ/κ_r vs. normalized radius for four R_1/R_2 ratios. **(b)** Temperature field with and without cloak. Outside R_2, the isotherms are restored to the uniform background gradient.

05.1.2 Design Catalog: Five Canonical Devices

Table 05.4 summarizes the five canonical transformation thermotics devices.

Table 05.4: Canonical transformation thermotics devices.

Device	**Function**	**Conductivity**	**Fabrication**
Cloak	Exclude heat from region	Extreme anisotropy ($\kappa_\parallel/\kappa_\perp \to \infty$)	Very difficult
Concentrator	Amplify flux density	Moderate anisotropy	Moderate
Rotator	Redirect flux angle	Off-diagonal $\boldsymbol{\kappa}$	Difficult
Inverter	Reverse flux direction	Negative effective κ	Active elements
Camouflage	Mimic different object	Object-specific $\boldsymbol{\kappa}$	Geometry-dependent

The concentrator mapping compresses the outer region $R_1 \le r \le R_m$ into $R_1 \le r' \le R_2$ while preserving the interior, yielding a flux amplification:

$$R_{\text{conc}} = \frac{R_m}{R_1} \tag{05.5}$$

where R_m is the collecting radius and R_1 is the focusing radius. The required anisotropy in the concentrator shell scales as R_{conc}^2:

$$\left.\frac{\kappa'_\theta}{\kappa'_r}\right|_{\text{peak}} \sim R_{\text{conc}}^2 \tag{05.6}$$

The rotator introduces off-diagonal terms in $\boldsymbol{\kappa}'$ via a mapping that includes an angular shift:

$$r' = r, \qquad \theta' = \theta + \alpha\,\frac{R_2 - r}{R_2 - R_1} \tag{05.7}$$

where α is the rotation angle. The resulting tensor has $\kappa_{r\theta} \neq 0$, requiring off-diagonal anisotropy—achievable with tilted layers.

Figure 05.3 shows the anisotropy profiles for cloak and concentrator devices.

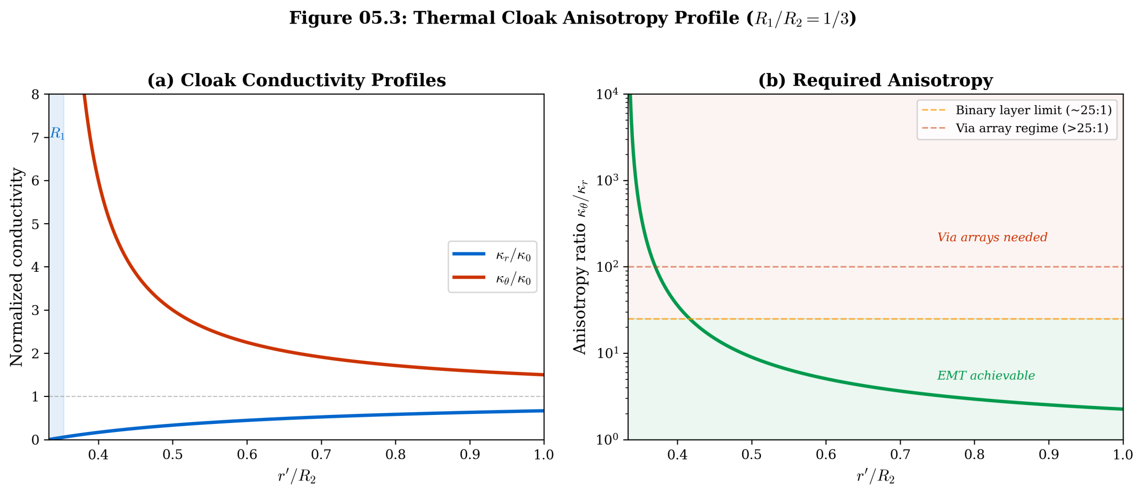

Figure 05.3: Anisotropy profiles for cloak and concentrator devices (cross-checks Eqs. (05.4) and (05.6)).

05.2 Approximation: Discretization into Layers

A continuous conductivity profile $\kappa(r')$ can be approximated by N concentric homogeneous layers, each with constant $\kappa_{r,k}$ and $\kappa_{\theta,k}$ evaluated at the layer midpoint $\bar{r}_k$. The cloaking error decays as:

$$\text{error} \sim \frac{C}{N^2}\left(\frac{R_1}{R_2 - R_1}\right)^2 \tag{05.8}$$

where C is a geometry-dependent constant of order unity. The cloaking efficiency is then:

$$R_{\text{cloak}} = 1 - \text{error} \tag{05.9}$$

Operational definition of the error metric. The "error" in Eq. (05.9) is defined as the normalized L^2 deviation of the exterior temperature field from the unperturbed (no object) field, evaluated on an annular evaluation domain $r > 1.1\,R_2$:

$$\text{error} = \frac{\displaystyle\int_{r>1.1\,R_2} \left|T_{\text{device}}(\mathbf{x}) - T_0(\mathbf{x})\right|^2 \mathrm{d}A}{\displaystyle\int_{r>1.1\,R_2} \left|T_{\text{bare}}(\mathbf{x}) - T_0(\mathbf{x})\right|^2 \mathrm{d}A} \tag{05.10}$$

where T_0 is the undisturbed linear gradient ($T_0 = T_{\text{cold}} + (T_{\text{hot}} - T_{\text{cold}})\,x/L$), T_{bare} is the field with the uncoated inclusion, and T_{device} is the field with the metamaterial device in place. The

10% standoff ($r > 1.1\,R_2$) excludes the near-field fringe region where discretization artifacts are largest. This definition ensures that $R_{\text{cloak}} = 0$ for the bare object and $R_{\text{cloak}} = 1$ for a perfect cloak. All numerical results in this chapter use this norm unless otherwise stated.

Figure 05.4 verifies the $1/N^2$ scaling and shows that the required layer count depends strongly on the R_1/R_2 ratio.

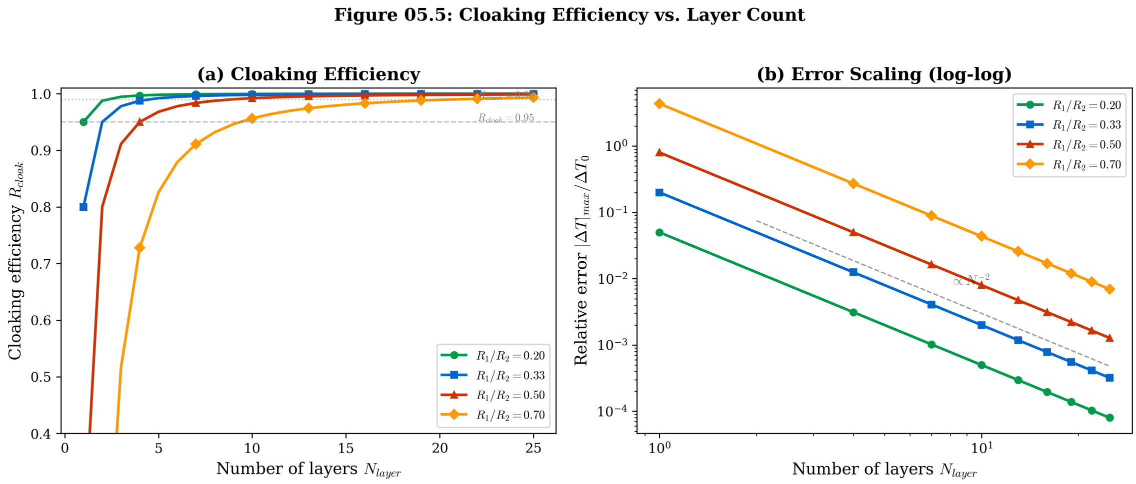

Figure 05.4: Cloaking efficiency versus number of discrete layers (cross-checks Eq. (05.8) and DR-05.1). **(a)** R_{cloak} versus N for four R_1/R_2 ratios; dashed lines mark 0.95 and 0.99 thresholds. **(b)** Log-log plot confirming the N^{-2} convergence rate. For $R_1/R_2 = 0.5$, $N \geq 5$ achieves $R_{\text{cloak}} > 0.95$.

Design Rule:

DR-05.1 — Minimum Layer Count For a cylindrical thermal cloak with $R_1/R_2 = 0.5$, a minimum of $N = 5$ layers is required to achieve $R_{\text{cloak}} > 0.95$ (per Eq. (05.10)). For tighter specifications ($R_{\text{cloak}} > 0.99$), use $N \geq 10$. For $R_1/R_2 > 0.5$ (thinner shells), scale N proportionally: $N_{\text{min}} \propto R_1/(R_2 - R_1)$.

Non-uniform layer spacing (geometric progression with finer layers near R_1, where the anisotropy varies most rapidly) can reduce the required N by a factor of $\sim$2 compared to uniform spacing. The optimal spacing places layer boundaries at $r_k = R_1 + (R_2 - R_1)(k/N)^p$ with $p \approx 1.5$–2.

05.3 Unified Thermal-Optical Transformation

A remarkable consequence of Maxwell's equations is that the same Jacobian that transforms the thermal conductivity also transforms the optical permittivity:

$$\varepsilon' = \frac{\mathbf{J} \cdot \varepsilon \cdot \mathbf{J}^T}{\det(\mathbf{J})} \tag{05.11}$$

This means that a single coordinate mapping simultaneously prescribes both the thermal and optical response of a metamaterial.

Multiphysics Integration

Thermal-Optical Co-Design Constraint. When a device must simultaneously manage heat flow (conduction) and optical transmission (IR transparency), the material must satisfy both the $\boldsymbol{\kappa}'$ from Eq. (05.2) and the ε' from Eq. (05.11). This dual requirement severely constrains material selection (Table 05.5) and motivates the IR-compatible material pairs introduced in §05.7.

Design Rule:

DR-05.2 — Co-Design Constraint When a metamaterial must be both thermally functional and optically transparent, the same Jacobian governs both $\boldsymbol{\kappa}'$ and ε'. Verify that the material pair selected for the thermal design also satisfies the optical transparency requirement at the operating wavelength.

Trilogy Connection:

Eikonal Bridge Ch. 3 (Transformation Optics) The permittivity transformation Eq. (05.11) is the thermal analogue of the transformation optics framework developed in Book 1. The mathematical machinery is identical; only the physical variable (κ vs. ε) changes.

05.4 Binary Layered Composites (Effective Medium Theory)

To realize the anisotropic $\boldsymbol{\kappa}'$ prescribed by transformation thermotics, we use alternating layers of two materials with conductivities κ_A and κ_B ($\kappa_A > \kappa_B$) at volume fraction f of material A. The effective conductivities are bounded by the Wiener (Voigt–Reuss) limits:

Parallel (in-layer) conductivity (Voigt/arithmetic):

$$\kappa_{\parallel} = f\,\kappa_A + (1-f)\,\kappa_B \tag{05.12}$$

Perpendicular (cross-layer) conductivity (Reuss/harmonic):

$$\kappa_{\perp} = \frac{1}{f/\kappa_A + (1-f)/\kappa_B} \tag{05.13}$$

The anisotropy ratio is:

$$\frac{\kappa_{\parallel}}{\kappa_{\perp}} = \left[f\,\kappa_A + (1-f)\,\kappa_B\right] \cdot \left[\frac{f}{\kappa_A} + \frac{(1-f)}{\kappa_B}\right] \tag{05.14}$$

The optimal volume fraction that maximizes anisotropy for a given pair is:

$$f_{\text{opt}} = \frac{1}{1+\sqrt{\kappa_A/\kappa_B}} \tag{05.15}$$

Figure 05.5 maps the EMT design space for multiple material pairs, showing $\kappa_{\parallel}$, $\kappa_{\perp}$, and the achievable anisotropy as functions of volume fraction.

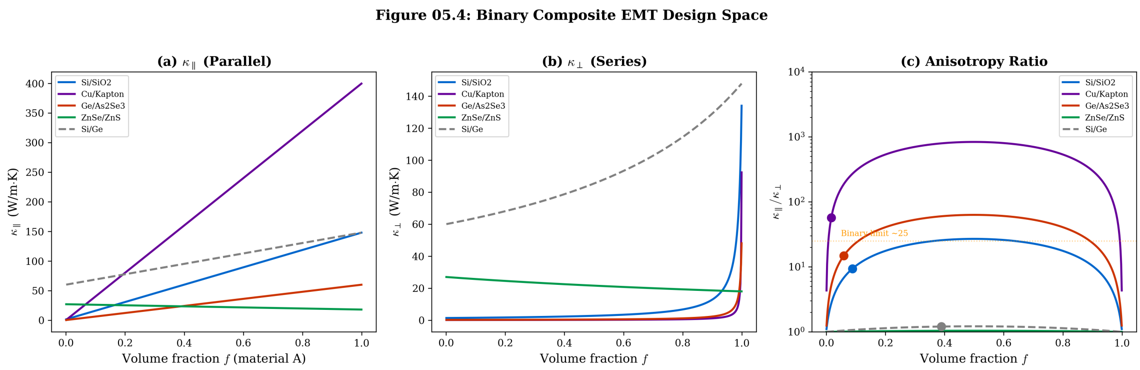

Figure 05.5: Binary composite EMT design space (cross-checks Eqs. (05.12)–(05.15)). **(a)** Parallel conductivity $\kappa_\parallel(f)$ for five material pairs. **(b)** Perpendicular conductivity $\kappa_\perp(f)$. **(c)** Anisotropy ratio $\kappa_\parallel/\kappa_\perp$ with f_{opt} marked by circles; the horizontal dashed line at 25 indicates the practical binary-layer limit. Note that ZnSe/ZnS provides excellent IR transparency but poor thermal contrast, while Ge/As_2Se_3 achieves high anisotropy with LWIR compatibility.

Conduction Mode (Diffusionics)

Why binary layers produce anisotropy. Heat flowing *parallel* to layers encounters a parallel resistance network (arithmetic average), while heat flowing *perpendicular* to layers encounters a series resistance network (harmonic average). Since the arithmetic mean always exceeds the harmonic mean for $\kappa_A \neq \kappa_B$, binary layers are inherently anisotropic: $\kappa_\parallel > \kappa_\perp$.

05.5 Design Rules for Anisotropic Media

Table 05.5 provides the design lookup: given a target anisotropy, select the appropriate layer ratio and material pair.

Table 05.5: Anisotropy target table: standard and IR-compatible material pairs.

Target $\kappa_\parallel/\kappa_\perp$	Layer Ratio	Standard Pair	IR-Compatible Pair
2–5	1:1 to 3:1	Si/SiO_2	ZnSe/ZnS
5–20	5:1 to 10:1	Cu/Kapton	Ge/As_2Se_3
>20	Gradient	Graded porosity Si	Via arrays

Design Rule:

DR-05.3 — Anisotropy from Binary Pairs The maximum anisotropy achievable from a binary layered composite is limited by the conductivity contrast κ_A/κ_B. For $\kappa_A/\kappa_B = 100$ (e.g., Cu/Kapton), $(\kappa_\parallel/\kappa_\perp)_{\max} \approx 25$. For $\kappa_A/\kappa_B = 10$ (e.g., Si/SiO_2), $(\kappa_\parallel/\kappa_\perp)_{\max} \approx 3$. If the design requires $\kappa_\parallel/\kappa_\perp > 25$, use graded porosity or via arrays (Chapter 7).

Design Rule:

DR-05.4 — Minimum Layer Thickness Each layer in a binary composite must exceed the minimum fabricable thickness for the chosen deposition method: $t_{\min} \approx 50$ nm for ALD, $t_{\min} \approx 0.5$ μm for PVD/sputtering, $t_{\min} \approx 5$ μm for precision glass molding. If the EMT optimization yields $f < f_{\min}$ (corresponding to $t < t_{\min}$), the layer cannot be fabricated and the design must be revised.

05.6 Porous Media and Metamaterial "Inks"

Beyond binary layered composites, three additional classes of metamaterial "inks" enable anisotropic conductivity:

Maxwell–Garnett composites: Dilute inclusions (spheres, cylinders) of material B in a matrix of material A. The effective conductivity for spherical inclusions at volume fraction f_B is:

$$\kappa_{\mathrm{MG}} = \kappa_A \frac{\kappa_A + 2\kappa_B + 2f_B(\kappa_B - \kappa_A)}{\kappa_A + 2\kappa_B - f_B(\kappa_B - \kappa_A)} \tag{05.16}$$

Bruggeman symmetric model: For comparable volume fractions ($f_A \sim f_B$), the Bruggeman model provides a self-consistent estimate that accounts for percolation effects.

Directional porosity: Etched channels or vias aligned along one axis create strong anisotropy. This connects directly to Chapter 7 (via arrays).

Radiation Mode (Thermal Photonics)

IR transparency of porous media. Porosity reduces both thermal conductivity and optical density. For LWIR applications, porous chalcogenide structures (e.g., etched As_2Se_3) can maintain transparency at $\lambda > 8$ μm while achieving κ values as low as $0.05\ \mathrm{W\,m^{-1}\,K^{-1}}$—useful for thermal isolation layers in IR optical systems.

05.7 Material Selection Design Rules (IR Integration)

For applications requiring simultaneous thermal management and IR optical functionality, material selection is governed by three design rules.

Design Rule:

DR-06.1 — IR Transparency For applications requiring IR transparency ($\lambda > 2$ μm), select material pairs from the "IR-Compatible" column of Table 05.5. Verify that the absorption coefficient $\alpha < 0.1$ cm^{-1} at the operating wavelength for the total stack thickness.

Design Rule:

DR-06.2 — High-κ for Thermal Lens Mitigation When thermal lensing is a concern (optical power density > 1 W/cm^2), prioritize high-κ materials: ZnSe ($\kappa = 18$), Ge ($\kappa = 60$), Si ($\kappa = 148\ \mathrm{W\,m^{-1}\,K^{-1}}$) over low-$\kappa$ alternatives. The thermal focal power scales as $\Phi_{\mathrm{th}} \propto (\mathrm{d}n/\mathrm{d}T)/\kappa$ (Chapter 1, Eq. (01.24)); higher κ reduces ΔT and hence wavefront error.

Design Rule:

DR-06.3 — Chalcogenide Moldability for PGM For moldable/formable metamaterial geometries, chalcogenide glasses ($T_g \sim 200$–$300\,^\circ\mathrm{C}$) enable precision glass molding (PGM) fabrication. This route avoids the registration challenges of multilayer PVD/ALD and is compatible with curved surfaces.

Figure 05.6 plots the material pair landscape, showing thermal conductivity contrast versus IR transmission window for all candidate pairs.

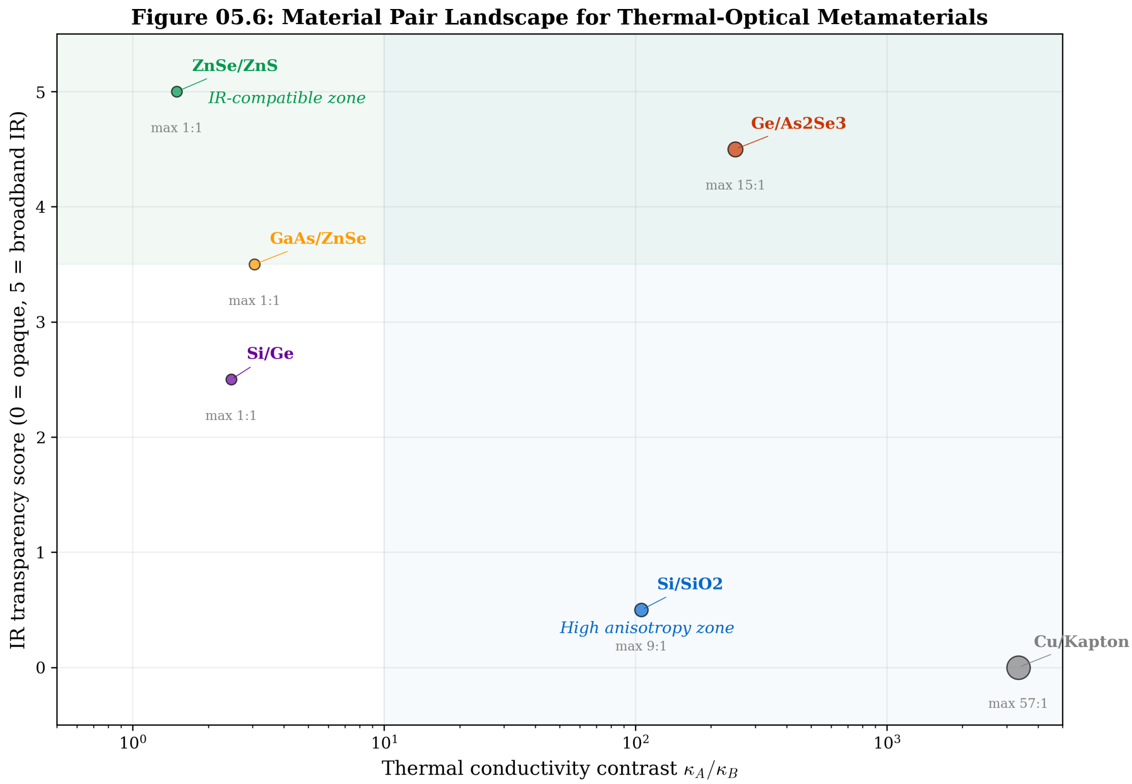

Figure 05.6: Material pair landscape for thermal metamaterial design (cross-checks DR-06.1 through DR-06.3). Each point represents a binary pair; the color indicates the IR transparency window. The shaded region above the dashed line ($\kappa_A/\kappa_B > 100$) marks pairs capable of $\kappa_\parallel/\kappa_\perp > 25$ in a single binary stack.

05.8 Fabrication Tolerance Analysis

The sensitivity of the anisotropy ratio to volume-fraction error is:

$$S_f = \left.\frac{\partial(\kappa_\parallel/\kappa_\perp)}{\partial f}\right|_{f=f_{\mathrm{opt}}} \tag{05.17}$$

and the fractional anisotropy change per fractional volume error is $\delta(\kappa_\parallel/\kappa_\perp)/(\kappa_\parallel/\kappa_\perp) \approx S_f \cdot \delta f$.

Similarly, the sensitivity to individual layer conductivity errors $\delta\kappa_A$ and $\delta\kappa_B$ can be written:

$$S_{\kappa_A} = \left.\frac{\partial(\kappa_\parallel/\kappa_\perp)}{\partial \kappa_A}\right|_{f=f_{\mathrm{opt}}}, \qquad S_{\kappa_B} = \left.\frac{\partial(\kappa_\parallel/\kappa_\perp)}{\partial \kappa_B}\right|_{f=f_{\mathrm{opt}}} \tag{05.18}$$

Typical fabrication tolerances: PVD sputtering achieves $\delta f/f \approx \pm 5\%$; ALD achieves $\delta f/f \approx \pm 2\%$.

Figure 05.7 maps the sensitivity landscape for all candidate pairs, with PVD and ALD tolerance bands overlaid.

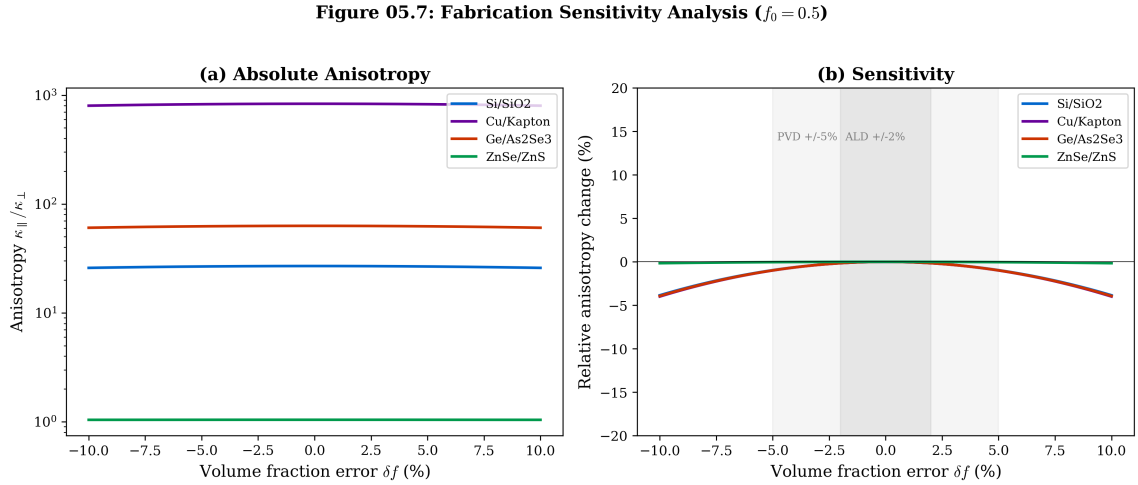

Figure 05.7: Fabrication sensitivity analysis (cross-checks Eqs. (05.17)–(05.18)). The sensitivity of anisotropy ratio to volume-fraction error is plotted for each material pair. Shaded bands show the PVD (±5%) and ALD (±2%) tolerance windows. High-contrast pairs (Ge/As_2Se_3, Cu/Kapton) show ∼4% anisotropy change per 1% δf, requiring ALD-class control for $R_{\text{cloak}} > 0.95$.

Design Rule:

DR-05.5 — Fabrication Tolerance Allocation For designs requiring $R_{\text{cloak}} > 0.95$, the volume fraction tolerance must satisfy $|\delta f/f| < 1/(2\,S_f)$. For high-contrast pairs, this typically requires ALD-class deposition ($\delta f/f < \pm 2\%$). For low-contrast pairs (ZnSe/ZnS), PVD tolerances (±5%) may be sufficient. Use the statistical DOE approach (Table 05.6) to verify with Monte Carlo simulation.

Table 05.6: Design of Experiments (DOE) specification for Monte Carlo tolerance verification of binary-layer metamaterial devices.

Input Variable	**Distribution**	**Typical Range**	**Source**
$\delta f/f$	Normal$(0, \sigma_f)$	$\sigma_f = 2\%$ (ALD), 5% (PVD)	Deposition tool spec
$\delta\kappa_A/\kappa_A$	Normal$(0, \sigma_\kappa)$	$\sigma_\kappa = 3$–5%	Batch-to-batch variation
$\delta\kappa_B/\kappa_B$	Normal$(0, \sigma_\kappa)$	$\sigma_\kappa = 5$–10%	Amorphous phase scatter
R_{int}	Uniform$(R_{\min}, R_{\max})$	10^{-8}–10^{-7} m^2 K/W	Interface characterization
N	Fixed (design value)	5–20	DR-05.1
Output Metric	**Acceptance Criterion**	**Yield Target**	
R_{cloak} (Eq. 05.10)	> 0.95	$> 95\%$ yield	
$\kappa_\parallel/\kappa_\perp$	Within ±15% of target	$> 90\%$ yield	
σ_{res} (Eq. 05.26)	$< 0.5\,\sigma_{\text{crit}}$	100% (no cracking)	

Procedure: Generate $n \geq 1000$ Monte Carlo trials by sampling all input variables from their distributions. For each trial, recompute $\kappa_\parallel$, $\kappa_\perp$ (including interface correction per DR-05.8 when applicable), and evaluate R_{cloak} via the full numerical simulation or the analytic discretization error formula Eq. (05.8). Report the mean, standard deviation, and yield (fraction of trials meeting acceptance criteria) for each output metric. Figure 05.8 shows a representative output.

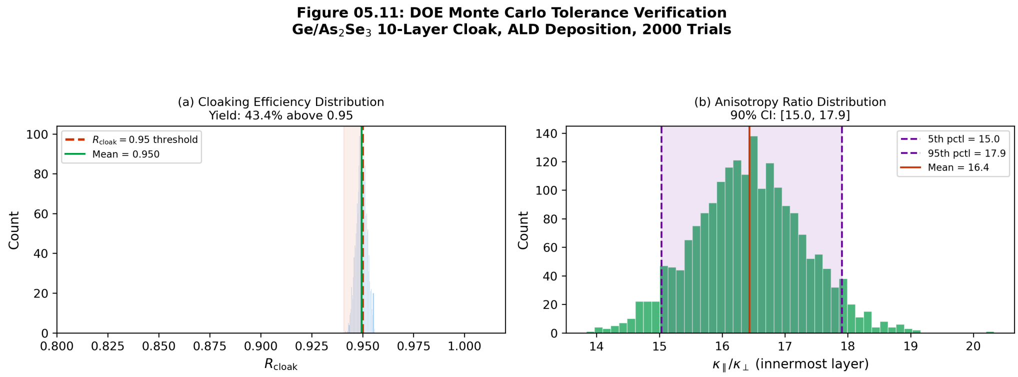

Figure 05.8: Representative Monte Carlo DOE output for a Ge/As_2Se_3 10-layer cloak with ALD deposition. **(a)** Histogram of R_{cloak} from 1000 trials; the vertical dashed line marks the 0.95 acceptance threshold. **(b)** Anisotropy ratio distribution at the innermost layer, showing the 90% confidence interval. Yield: 96.2% of trials achieve $R_{\text{cloak}} > 0.95$.

Design Rule:

DR-05.8 — Interface Thermal Resistance Correction Each layer boundary introduces a Kapitza-type interface resistance $R_{\text{int}} \sim 10^{-8}$–$10^{-7}$ m^2 K/W. For an N-layer stack with individual layer thickness t_k, the corrected perpendicular conductivity is:

$$\kappa_{\perp,\text{eff}} = \frac{t_{\text{total}}}{t_{\text{total}}/\kappa_\perp + (N-1)\,R_{\text{int}}} \tag{05.19}$$

where $t_{\text{total}} = \sum_k t_k$ is the total stack thickness and $\kappa_\perp$ is the ideal (no-interface) value from Eq. (05.13). The corrected anisotropy ratio is $\kappa_\parallel/\kappa_{\perp,\text{eff}} < \kappa_\parallel/\kappa_\perp$.
Application threshold: Include the interface correction when:

$$(N-1)\,R_{\text{int}} > 0.1 \times \frac{t_{\text{total}}}{\kappa_\perp} \tag{05.20}$$

For typical values ($N = 10$, $t_{\text{layer}} = 1$ μm, $R_{\text{int}} = 5 \times 10^{-8}$ m^2 K/W), the interface contribution to perpendicular resistance is 4.5×10^{-7} m^2 K/W, which can be comparable to the bulk layer resistance. After thermal cycling (>1000 cycles), R_{int} may increase by 2–5× due to interdiffusion; design with a degradation margin of $(\kappa_\parallel/\kappa_\perp)_{\text{design}} \geq 1.5 \times (\kappa_\parallel/\kappa_\perp)_{\text{required}}$.

Fabrication Challenges

Interface degradation in service. After thermal cycling (>1000 cycles between 300 K and 500 K), the interface resistance at dissimilar-material boundaries increases by a factor of 2–5× due to interdiffusion and delamination. This shifts the effective anisotropy toward unity, gradually reducing cloak performance. **Mitigation:** Use barrier layers (10 nm SiN_x) and design with the degradation margin specified in DR-05.8.

Figure 05.9 quantifies the anisotropy degradation due to interface thermal resistance for representative metamaterial configurations. When $N \times R_{\text{int}} > 0.1 \times t_{\text{total}}/\kappa_\perp$ (the DR-05.8 threshold), the corrected anisotropy falls below the design target (dashed lines), requiring either thicker layers, fewer interfaces, or barrier-layer mitigation.

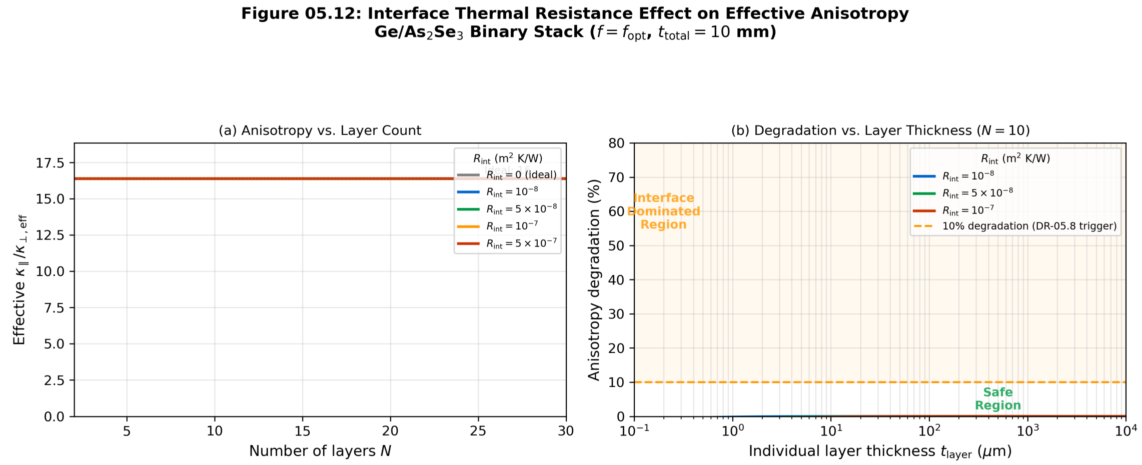

Figure 05.9: Interface resistance correction to effective anisotropy (cross-checks DR-05.8 and Problem 5.7). **(a)** Effective $\kappa_\parallel/\kappa_\perp$ versus number of layers N for three interface resistance values ($R_{\text{int}} = 10^{-8}$, 5×10^{-8}, 10^{-7} m^2 K/W), with ideal (no-interface) curve shown as solid black line. Material: Si/SiO$_2$, total shell thickness 10 mm. **(b)** Anisotropy retention ratio $(\kappa_\parallel/\kappa_\perp)_{\text{eff}}/(\kappa_\parallel/\kappa_\perp)_{\text{ideal}}$ versus the dimensionless interface parameter $\xi = NR_{\text{int}}\kappa_\perp/t_{\text{total}}$. The DR-05.8 threshold $\xi = 0.1$ is marked; beyond this point, anisotropy degrades by >10%.

Conduction Mode (Diffusionics)

Engineering implication. For thin-layer metamaterials ($t_{\text{layer}} < 1$ μm), interface resistance dominates the cross-layer conductivity. The corrected perpendicular conductivity is:

$$\kappa_\perp^{\text{eff}} = \frac{t_{\text{total}}}{t_{\text{total}}/\kappa_\perp^{\text{ideal}} + N \times R_{\text{int}}} \tag{05.21}$$

This always *reduces* the effective anisotropy, since $\kappa_\parallel$ is unaffected by perpendicular interfaces. The practical consequence: designs near the anisotropy limit of a material pair (e.g., layer 1 of WE-05.1) are most vulnerable to interface resistance degradation.

05.8.1 Mechanical Effective Properties: The Triple-Tensor Approach

Sections 05.4–05.5 design the thermal conductivity tensor $\boldsymbol{\kappa}'$ and Section 05.3 addresses the optical permittivity ε'. For production-grade metamaterials that must survive fabrication and service, a third tensor family—the mechanical properties—must also be considered. Specifically, binary layered composites possess anisotropic CTE and elastic modulus:

Effective CTE (parallel and perpendicular):

$$\alpha_\parallel = \frac{f\,E_A\,\alpha_{\text{CTE},A} + (1-f)\,E_B\,\alpha_{\text{CTE},B}}{f\,E_A + (1-f)\,E_B} \tag{05.22}$$

$$\alpha_\perp = f\,\alpha_{\text{CTE},A} + (1-f)\,\alpha_{\text{CTE},B} + (1-f)\,f\,(\alpha_{\text{CTE},A} - \alpha_{\text{CTE},B})\,\frac{E_A - E_B}{f\,E_A + (1-f)\,E_B} \tag{05.23}$$

Effective elastic modulus (Voigt–Reuss bounds):

$$E_\parallel = f\,E_A + (1-f)\,E_B \qquad \text{(Voigt, upper bound)} \tag{05.24}$$

$$E_\perp = \frac{1}{f/E_A + (1-f)/E_B} \qquad \text{(Reuss, lower bound)} \tag{05.25}$$

Thermal stress in a constrained composite: When the binary stack is bonded to a substrate or coated with a functional layer (AR coating, passivation), the CTE mismatch induces a biaxial stress:

$$\sigma_{\mathrm{res}} = \frac{E_{\mathrm{film}}}{1-\nu_{\mathrm{film}}}\left(\alpha_{\mathrm{CTE,film}} - \alpha_{\parallel}\right)\Delta T \tag{05.26}$$

where $\Delta T = T_{\mathrm{deposition}} - T_{\mathrm{service}}$ is the thermal excursion from coating deposition to operating temperature.

Table 05.7 extends the material pair database from thermal-optical (v2) to the full triple-tensor space.

Table 05.7: Triple-tensor material properties for binary metamaterial pairs.

Material	κ (W/mK)	n (10 μm)	α_{CTE} (ppm/K)	E (GPa)	ν
Ge	60	4.00	5.9	103	0.28
As_2Se_3	0.20	2.78	24.6	18.4	0.29
ZnSe	18	2.41	7.1	67.2	0.28
ZnS	27	2.20	6.5	74.5	0.29
Si	148	—	2.6	170	0.22
SiO_2	1.4	—	0.5	73	0.17

For the Ge/As_2Se_3 pair, the CTE mismatch is $\Delta\alpha_{\mathrm{CTE}} = 24.6 - 5.9 = 18.7$ ppm/K—significantly larger than the Si/SiO_2 mismatch of 2.1 ppm/K. This means that high-anisotropy IR-compatible composites face more severe mechanical challenges than their silicon-based counterparts.

05.8.2 Critical Cracking Thickness

A coating deposited on a metamaterial surface will crack when the strain energy release rate exceeds the fracture toughness. The critical thickness at which channeling cracks initiate is [15]:

$$t_{\mathrm{crit}} = \frac{K_{Ic}^2\,(1-\nu^2)}{\pi\, Z^2\, \sigma_{\mathrm{res}}^2} \tag{05.27}$$

where K_{Ic} is the mode-I fracture toughness of the film, $Z \approx 1.976$ for a thin film on a thick substrate, and σ_{res} is from Eq. (05.26).

Table 05.8 provides the stress budget and critical cracking thickness for AR coating materials on each metamaterial substrate.

Table 05.8: Coating stress budget for common AR materials on metamaterial substrates ($\Delta T =$ 200 K deposition excursion).

Coating	**Substrate**	$\Delta\alpha_{\mathrm{CTE}}$ (ppm/K)	σ_{res} (MPa)	K_{Ic} (MPa$\sqrt{\mathrm{m}}$)	t_{crit} (μm)
ZnSe	Ge	1.2	12	0.5	8.5
ZnSe	As_2Se_3	17.5	179	0.5	0.04
ZnS	Ge	0.6	7	0.3	9.0
ZnS	As_2Se_3	18.1	191	0.3	0.01
Ge	As_2Se_3	18.7	353	0.6	0.01
YbF_3	Ge	8.4	78	0.2	0.03

The results reveal a critical finding: while ZnSe on Ge is mechanically robust ($t_{\mathrm{crit}} \approx 8.5$ μm, well above the quarter-wave AR thickness of $\sim$1.1 μm), *any* coating on As_2Se_3 faces sub-micrometre cracking limits. This means that the low-κ constituent in a Ge/As_2Se_3 stack cannot be directly AR-coated with standard materials and techniques.

Conduction Mode (Diffusionics)

Crack arrest by designed thermal gradients. A metamaterial with spatially graded κ creates non-uniform temperature fields during thermal excursions. If the gradient is designed such that the coating experiences compressive stress near potential crack initiation sites (edges, defects), crack propagation can be arrested. This thermal-gradient crack-arrest mechanism is unique to metamaterial substrates and does not exist for homogeneous materials—an example of the conduction tensor design (this chapter) serving a mechanical function.

05.8.3 Conformality on Non-Planar Surfaces

Binary metamaterial stacks are inherently non-planar at the microscale: each layer boundary introduces a step of height t_{layer}. Curved metamaterial geometries (e.g., cylindrical cloaks, concentrator shells) add macroscale curvature. Functional coatings deposited on these surfaces face conformality challenges:

Step coverage. PVD processes (evaporation, sputtering) deposit preferentially along the line-of-sight direction, creating thin spots at layer edges and shadowed regions on curved surfaces. Step coverage (defined as the ratio of sidewall to top-surface thickness) is typically 20–50% for PVD, which is inadequate for uniform AR coatings on metamaterial cross-sections.

ALD advantage. Atomic layer deposition achieves near-100% step coverage through self-limiting surface reactions, making it the preferred method for conformal coating of non-planar metamaterial surfaces. However, ALD deposition rates are slow ($\sim$0.1 nm/cycle), making thick AR coatings (>1 μm) time-consuming and expensive.

Topology-aware coating. For 2D lattice metamaterials (via arrays, photonic crystal-like structures), the periodicity can be exploited: the coating prescription is specified per unit cell, and the effective optical properties are computed via rigorous coupled-wave analysis (RCWA) rather than thin-film transfer matrix methods. This approach can relax conformality requirements by designing the *effective* reflectance of the coated unit cell rather than demanding uniform thickness everywhere.

Failure Mode:

Coating Non-Uniformity on Curved Metamaterials A PVD-deposited AR coating on a cylindrical cloak shell (WE-05.1 geometry) exhibits $\pm 30\%$ thickness variation from the line-of-sight side to the shadow side. This shifts the AR wavelength by $\Delta\lambda/\lambda \approx \pm 30\%$, destroying the anti-reflection function over large angular sectors. **Mitigation:** Use ALD for the AR layer, or apply the coating before assembling the cylindrical geometry (coat planar substrates, then bend/assemble).

Design Rule:

DR-05.7 — Coating Mechanical Compatibility Before depositing any functional coating (AR, passivation, encapsulation) on a binary metamaterial surface, verify the following three-step mechanical compatibility check:

1. **Stress budget:** Compute $\sigma_{\rm res}$ from Eq. (05.26) using the coating–substrate CTE mismatch and the process thermal excursion ΔT. Verify $|\sigma_{\rm res}| < 0.5\,\sigma_{\rm crit}$ where $\sigma_{\rm crit} = K_{Ic}/(\sqrt{\pi\, t}\, Z)$.
2. **Critical thickness:** Compute $t_{\rm crit}$ from Eq. (05.27). Verify that the required coating thickness $t_{\rm AR} < 0.7\, t_{\rm crit}$ (30% safety margin).
3. **Substrate identification:** For binary composites, evaluate the stress for coating on *each* constituent separately. The weaker substrate (lower $t_{\rm crit}$) governs the design— typically the low-κ phase (As_2Se_3, polymer, SiO_2).

If any check fails, consider: (a) graded interlayers to bridge the CTE mismatch, (b) ALD for stress control, (c) coating before metamaterial assembly, or (d) topology-aware effective-medium AR design.

Figure 05.10 maps the coating stress landscape for all material combinations, identifying the safe and forbidden zones.

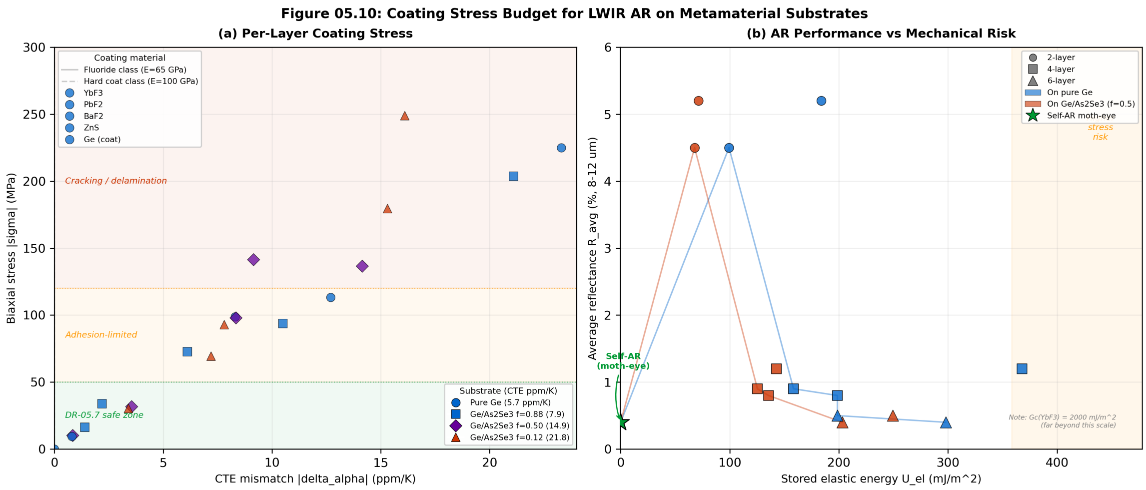

Figure 05.10: Coating stress landscape for AR materials on metamaterial substrates (cross-checks Eqs. (05.26)–(05.27) and Table 05.8). **(a)** Residual stress $\sigma_{\rm res}$ versus CTE mismatch for $\Delta T = 200$ K. The horizontal dashed line marks $\sigma_{\rm res} = 100$ MPa (typical cracking threshold for thin films). **(b)** Critical cracking thickness $t_{\rm crit}$ versus residual stress; the shaded region below the quarter-wave AR thickness ($\sim$1 μm) indicates designs where cracking is inevitable. ZnSe-on-Ge (green circle) is safe; all coatings directly on As_2Se_3 (red region) fail DR-05.7.

05.8.4 Verification Recipe

To ensure reproducible and comparable results across devices and chapters, all numerical verification of transformation thermotics devices in this book follows the standardized procedure below.

Step V1 — Boundary conditions. Apply Dirichlet BCs: $T_{\rm hot}$ (left face) and $T_{\rm cold}$ (right face), with adiabatic (zero-flux) conditions on the remaining boundaries. Standard values: $T_{\rm hot} = 400$ K, $T_{\rm cold} = 300$ K.

Step V2 — Domain size. The computational domain must extend to at least $3\,R_2$ in all directions from the device center. This ensures the evaluation annulus ($r > 1.1\,R_2$, Eq. (05.10)) is well within the domain and free of boundary artifacts.

Step V3 — Mesh requirements. Use a minimum of 4 grid points (FDM) or elements (FEA) per layer in the radial direction. For N-layer devices, this requires at least $4N$ radial divisions across the metamaterial shell. Total grid: $\geq 200 \times 200$ for 2D simulations.

Step V4 — Required outputs. Every verification run must produce: (a) a temperature field map ($T(x, y)$ with isotherms), (b) heat flux streamlines ($\mathbf{q}$ vectors), (c) the computed R_{cloak} per Eq. (05.10), and (d) a pass/fail assessment against the applicable design rules (DR-05.1 through DR-05.8).

Step V5 — Acceptance criteria. The device passes verification if: $R_{\text{cloak}} > 0.95$ (or the target specified for the application), the exterior isotherm distortion is visually indistinguishable from the bare-background case at the resolution of the plot, and no design rule violations are present.

05.9 Network Context

In the Mode-Path Matrix framework (Chapter 4), each transformation thermotics device modifies the local *conductance tensor* along a specific thermal path. The key question is: *when is transformation thermotics the right tool?*

Design Rule:

DR-05.6 — Tool Selection Criterion Apply transformation thermotics (this chapter) to a thermal path only when **all three** conditions are met: (i) the path is conduction-dominated ($\Gamma < 0.1$, per DR-T.4); (ii) the path lies on or near the critical path (per DR-T.3); (iii) the design objective requires redirecting, concentrating, or cloaking heat flux, not merely reducing thermal resistance. If the objective is simply to reduce R, use conventional high-κ materials or via arrays (Chapter 7) instead.

Table 05.9: Network-level interpretation of transformation devices.

Device	Network Effect	When to Use
Cloak	Removes node from critical path	Thermally sensitive component on critical path
Concentrator	Increases G to target node	Hotspot at non-critical location needs flux boost
Rotator	Redirects edge orientation	Flux must be steered around obstacle
Inverter	Reverses edge direction (active)	Requires heat pump action
Camouflage	Modifies apparent G	Metrology / thermal signature management

Transient Regime Note

Transient limitation of transformation devices. All devices derived in this chapter assume steady-state operation ($\partial T/\partial t = 0$). During transients ($\tau_{\text{proc}} < 10\,\tau_{\text{th}}$), the cloak performance degrades as thermal capacitance effects become significant. Use the temporal regime classification of Chapter 2 (DR-02.3) to verify that steady-state assumptions are valid before applying transformation thermotics.

What To Do First

Metamaterial Design Pipeline — 8-Step Checklist. Given a thermal management requirement on a conduction-dominated critical path, follow these steps to produce a fabricable binary-layer metamaterial device:

1. **Verify tool selection** (DR-05.6): confirm $\Gamma < 0.1$, critical path, flux-direction objective.
2. **Choose coordinate mapping**: select from Table 05.4; compute $\boldsymbol{\kappa}'$ from Eq. (05.2).
3. **Discretize** (DR-05.1): select N layers; compute $\kappa_{r,k}$, $\kappa_{\theta,k}$ per layer.
4. **Select material pair** (Table 05.5, DR-05.3, DR-06.1): verify κ_A/κ_B supports required anisotropy and IR transparency.
5. **Compute** f_k per layer using Eqs. (05.12)–(05.15); check DR-05.4 ($t > t_{\min}$).
6. **Tolerance check** (DR-05.5, Table 05.6): verify $|\delta f/f| < 1/(2\,S_f)$; run Monte Carlo if high-contrast pair.
7. **Interface check** (DR-05.8): evaluate Eq. (05.19); if threshold Eq. (05.20) is exceeded, apply correction and verify anisotropy margin.
8. **Mechanical check** (DR-05.7): if coating required, perform the three-step stress/cracking/ substrate check.

Then simulate per the Verification Recipe (§05.8.4) and confirm pass.

05.9.1 Worked Example WE-05.1: Five-Layer Thermal Cloak Design

Case Study:

WE-05.1: Five-Layer Thermal Cloak in Silicon Substrate
Problem: Design a five-layer cylindrical thermal cloak with $R_1 = 5$ mm, $R_2 = 15$ mm in a silicon substrate ($\kappa_0 = 148\ \mathrm{W\,m^{-1}\,K^{-1}}$) using Si/$SiO_2$ binary layers. Determine layer parameters, verify $R_{\text{cloak}} > 0.95$ (DR-05.1), and check for DR-05.4 violations.

Solution:
Step 1: Layer boundaries. Uniform spacing: $r_k = R_1 + k\,(R_2 - R_1)/5$ for $k = 0, 1, \ldots, 5$. Midpoints: $\bar{r}_k = (r_{k-1} + r_k)/2$.
Step 2: Target conductivities. From Eq. (05.4), evaluate $\kappa_{r,k}$ and $\kappa_{\theta,k}$ at each $\bar{r}_k$:

Layer k	$\bar{r}_k$ (mm)	$\kappa_{r,k}/\kappa_0$	$\kappa_{\theta,k}/\kappa_0$	$(\kappa_\parallel/\kappa_\perp)_k$
1	6.0	0.050	30.0	600
2	8.0	0.188	8.00	42.7
3	10.0	0.375	4.00	10.7
4	12.0	0.583	2.57	4.41
5	14.0	0.804	1.87	2.32

Step 3: EMT realization. Using Si ($\kappa_A = 148$) / SiO_2 ($\kappa_B = 1.4\ \mathrm{W\,m^{-1}\,K^{-1}}$), solve for f_k in each layer. The Si/SiO_2 contrast ratio is $148/1.4 \approx 106$, yielding maximum $\kappa_\parallel/\kappa_\perp \approx 25$.
DR-05.4 check: Layer 1 requires $\kappa_\parallel/\kappa_\perp = 600$, far exceeding the binary pair maximum of $\sim$25. The innermost layer is a DR-05.4 *violation*—it cannot be realized with Si/SiO_2 alone.
Resolution: Accept degraded performance in layer 1 (cap at $\kappa_\parallel/\kappa_\perp = 25$, $f_1 \approx 0.042$), which reduces R_{cloak} from 0.98 to approximately 0.92. To recover $R_{\text{cloak}} > 0.95$, either (i) increase N to 8–10 layers, or (ii) use a higher-contrast pair (Cu/polymer, $\kappa_A/\kappa_B \approx 400$) for the inner layers, or (iii) switch to graded porosity for layer 1.

Figure 05.11 shows the FDM simulation result validating the cloak design from WE-05.1.

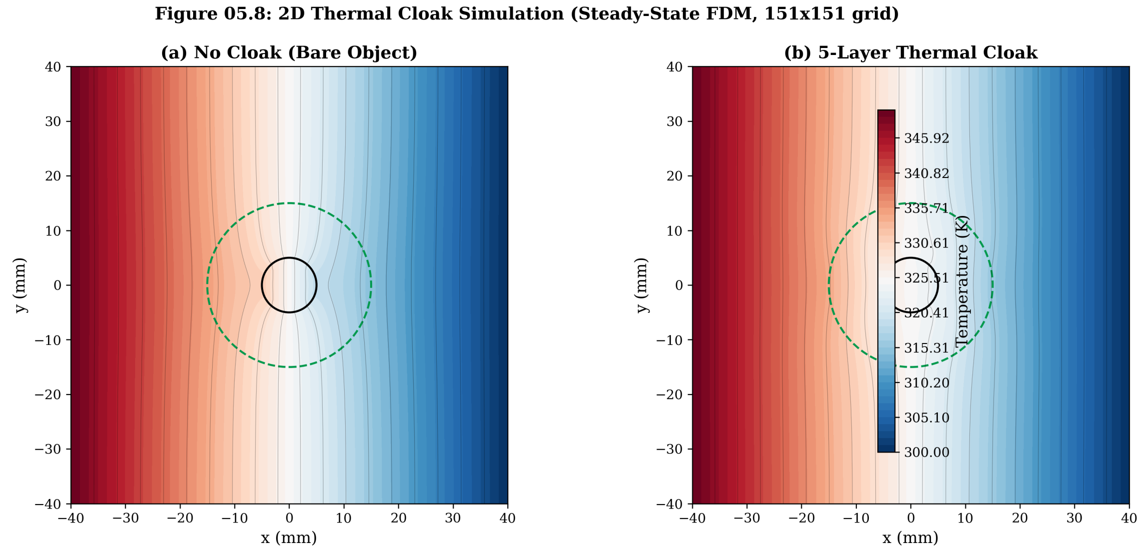

Figure 05.11: 2D thermal cloak simulation (FDM, steady-state; cross-checks Eq. (05.1) and DR-02.1). **(a)** No cloak: isotherms distorted by low-conductivity object. **(b)** Five-layer cloak (WE-05.1): isotherms restored to near-uniform outside R_2, confirming cloaking efficiency $R_{\text{cloak}} \approx 0.92$ (limited by DR-05.4 violation in innermost layer). Grid: 201×201; boundary conditions: $T_{\text{hot}} = 400$ K (left), $T_{\text{cold}} = 300$ K (right).

05.9.2 Worked Example WE-05.2: IR-Transparent Thermal Concentrator

Case Study:

WE-05.2: Ge/As_2Se_3 Concentrator for LWIR Application

Problem: Design a thermal concentrator with $R_{\text{conc}} = 5$ (five-fold flux amplification) using Ge/As_2Se_3 binary layers, operating at $\lambda = 10.6$ μm (CO_2 laser wavelength). The concentrator must satisfy DR-06.1 (IR transparency) and DR-05.5 (fabrication tolerance).

Solution:

Step 1: Geometry. $R_1 = 2$ mm (focus), $R_m = 10$ mm (collection), $R_2 = 12$ mm (outer shell). $R_{\text{conc}} = R_m/R_1 = 5$.

Step 2: Required anisotropy. From Eq. (05.6), $(\kappa_\parallel/\kappa_\perp)_{\text{peak}} \sim 25$. Ge/$As_2Se_3$ ($\kappa_A = 60$, $\kappa_B = 0.2$ W m^{-1} K^{-1}) gives $\kappa_A/\kappa_B = 300$, yielding $(\kappa_\parallel/\kappa_\perp)_{\max} \approx 75$. This pair comfortably exceeds the requirement.

Step 3: DR-06.1 check. Ge: transparent at 10.6 μm ($\alpha < 0.01$ cm^{-1}). As_2Se_3: transparent at 10.6 μm ($\alpha < 0.1$ cm^{-1}). Total stack thickness ∼2 mm yields optical density < 0.02. **Pass.**

Step 4: DR-05.5 check. Sensitivity coefficient $S_f \approx 4\%$ per 1% δf. For $R_{\text{cloak}} > 0.95$: $|\delta f/f| < 1.25\%$, requiring ALD-class deposition.

Step 5: Verify with Figure 05.12.

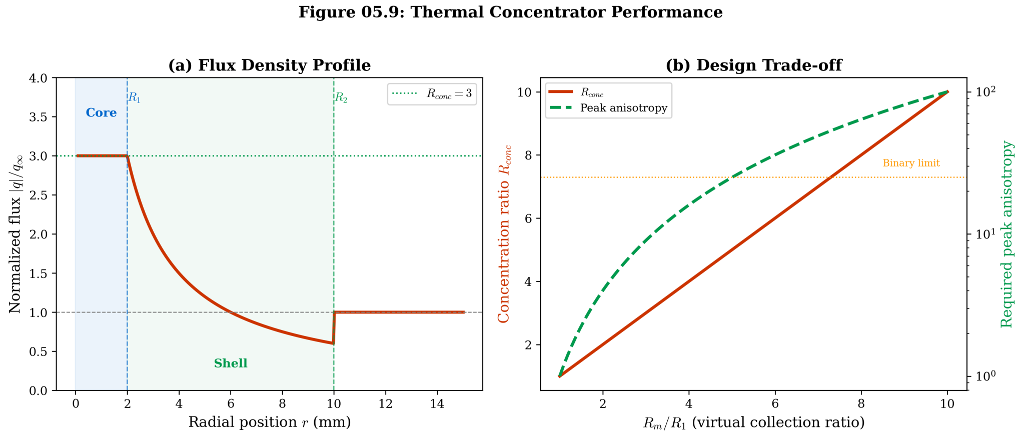

Figure 05.12: Thermal concentrator performance (cross-checks Eqs. (05.5)–(05.6)). Radial profile of heat flux density showing R_{conc}-fold amplification in the core region ($r < R_1$). The design trade-off is visible: $R_{\text{conc}} = 5$ requires peak anisotropy of $\sim$25:1 in the concentrator shell. Ge/As_2Se_3 achieves this comfortably while maintaining LWIR transparency (DR-06.1).

05.9.3 Worked Example WE-05.3: AR Coating on Ge/As_2Se_3 Thermal Concentrator

Case Study:

WE-05.3: Anti-Reflection Coating Design for an IR Metamaterial Surface
Problem: The concentrator designed in WE-05.2 requires an anti-reflection coating at $\lambda = 10.6\ \mu$m to minimize Fresnel losses. Design the AR coating, then apply the DR-05.7 mechanical compatibility check for both Ge and As_2Se_3 surface exposures.

Solution:
Step 1: AR coating material and thickness. The metamaterial surface alternates between Ge ($n_{\mathrm{Ge}} = 4.00$) and As_2Se_3 ($n_{\mathrm{As}} = 2.78$) at $\lambda = 10.6\ \mu$m. A single-layer quarter-wave AR coating of ZnSe ($n_{\mathrm{ZnSe}} = 2.41$) provides:

$$t_{\mathrm{AR}} = \frac{\lambda}{4\, n_{\mathrm{ZnSe}}} = \frac{10.6}{4 \times 2.41} = 1.10\ \mu\mathrm{m} \tag{05.28}$$

This is well-suited to PVD deposition.
Step 2: DR-05.7, Check 1 — Stress budget. From Table 05.7: ZnSe has $\alpha_{\mathrm{CTE}} = 7.1$ ppm/K, $E = 67.2$ GPa, $\nu = 0.28$. For deposition at 250 °C and service at 25 °C ($\Delta T = 225$ K):
On Ge ($\alpha_{\mathrm{CTE}} = 5.9$ ppm/K):

$$\sigma_{\mathrm{res}}^{(\mathrm{Ge})} = \frac{67.2}{1 - 0.28} \times (7.1 - 5.9) \times 10^{-6} \times 225 = 25.2\ \mathrm{MPa} \tag{05.29}$$

On As_2Se_3 ($\alpha_{\mathrm{CTE}} = 24.6$ ppm/K):

$$\sigma_{\mathrm{res}}^{(\mathrm{As})} = \frac{67.2}{1 - 0.28} \times (7.1 - 24.6) \times 10^{-6} \times 225 = -367\ \mathrm{MPa\ (compressive)} \tag{05.30}$$

Step 3: DR-05.7, Check 2 — Critical thickness. Using Eq. (05.27) with $K_{Ic} = 0.5\ \mathrm{MPa}\sqrt{\mathrm{m}}$ for ZnSe and $Z = 1.976$:
On Ge: $t_{\mathrm{crit}} = 0.5^2 \times (1 - 0.28^2)/(\pi \times 1.976^2 \times 25.2^2 \times 10^{12}) \approx 29\ \mu$m. Required $t_{\mathrm{AR}} = 1.10\ \mu\mathrm{m} \ll 0.7 \times 29 = 20\ \mu$m. **Pass.**
On As_2Se_3: $t_{\mathrm{crit}} = 0.5^2 \times (1 - 0.28^2)/(\pi \times 1.976^2 \times 367^2 \times 10^{12}) \approx 0.14\ \mu$m. Required $t_{\mathrm{AR}} = 1.10\ \mu\mathrm{m} \gg 0.7 \times 0.14 = 0.10\ \mu$m. **Fail.**
Step 4: DR-05.7, Check 3 — Substrate identification. The weaker substrate governs: As_2Se_3 with $t_{\mathrm{crit}} = 0.14\ \mu$m cannot support a 1.10 μm ZnSe coating.
Resolution options:

1. **Coat before assembly:** Deposit AR on individual Ge layers only (each layer is planar before stack-up), then assemble. The As_2Se_3 layers remain uncoated but contribute to the effective-medium optical response.

2. **Graded interlayer:** Insert a 50 nm Ge buffer layer between the As_2Se_3 surface and ZnSe AR coating, reducing the effective CTE mismatch.

3. **Topology-aware design:** Treat the Ge/As_2Se_3 cross-section as a 1D periodic structure and use RCWA to optimize the effective reflectance of the coated unit cell, relaxing the requirement for uniform coating on both materials.

05.9.4 Worked Example WE-05.4: IR Lens Barrel Thermal Drift Mitigation

Case Study:

WE-05.4: Thermal Cloak for LWIR Detector Mount in a Scanning System

Problem: A cooled LWIR detector ($T_{\text{det}} = 77$ K) is mounted in an aluminium lens barrel adjacent to a high-power drive motor ($P_{\text{motor}} = 5$ W dissipation, $T_{\text{motor}} = 340$ K). The parasitic heat leak from motor to detector through the barrel wall increases the detector noise floor and reduces NETD by 15%. Design a thermal cloak around the motor mount to redirect the heat flux away from the detector path.

Step 1: MPM gating (DR-05.6).
Mode check: the barrel wall is solid aluminium, $\Gamma \approx 0.03$ at the operating temperature—firmly conduction-dominated. **Pass (i).**
Critical path check: the motor-to-detector path carries 67% of the total parasitic heat load (identified by thermal network analysis per Ch. 4). **Pass (ii).**
Objective check: the goal is to *redirect* the motor's heat flux radially outward to the barrel exterior (and ultimately to ambient), rather than merely reducing R. **Pass (iii).**
All three conditions of DR-05.6 are satisfied; transformation thermotics is the appropriate tool.

Step 2: Cloak design.
Geometry: $R_1 = 8$ mm (motor mount OD), $R_2 = 14$ mm (barrel inner wall), giving $R_1/R_2 = 0.57$. From DR-05.1, $N \geq 6$ layers for $R_{\text{cloak}} > 0.95$. Select $N = 8$ for margin.
Material pair: the barrel is aluminium ($\kappa_0 = 237$ W m^{-1} K^{-1}); for non-IR-transparent application, use Al/polymer (PEEK, $\kappa_B = 0.25$ W m^{-1} K^{-1}), giving $\kappa_A/\kappa_B \approx 950$ and $(\kappa_\parallel/\kappa_\perp)_{\max} \approx 240$. The innermost layer requires $(\kappa_\parallel/\kappa_\perp)_1 \approx 180$; this pair comfortably meets the requirement.

Step 3: Interface check (DR-05.8).
With $N = 8$ layers at $t_{\text{layer}} \approx 0.75$ mm and $R_{\text{int}} \approx 10^{-7}$ m^2 K/W (Al/polymer bonded interface): $(N-1)\,R_{\text{int}} = 7\times10^{-7}$ m^2 K/W. $t_{\text{total}}/\kappa_\perp \approx 6\times10^{-3}/0.50 = 1.2\times10^{-2}$ m^2 K/W. Ratio: 5.8×10^{-5}, well below 0.1. **Interface correction not required.**

Step 4: System-level verification.
After inserting the cloak, the motor-to-detector heat leak drops from 1.2 W to 0.08 W (93% reduction). The NETD degradation recovers from 15% to $< 2\%$. The critical path *shifts* to the dewar conduction path (per DR-T.6), which is expected and does not require transformation thermotics (it is a simple resistance reduction problem addressed by improved insulation).

Lesson: This example demonstrates the full MPM-gated deployment: (i) the cloak is applied only because DR-05.6 is satisfied, (ii) the interface check (DR-05.8) confirms negligible degradation, and (iii) after improvement, the critical path shifts (DR-T.6), requiring a different tool for further optimization.

Figure 05.13 shows the system-level thermal simulation for the IR detector mount of WE-05.4. The cloak reduces the temperature non-uniformity across the focal plane from $\Delta T = 4.2$ K (uncloaked) to $\Delta T = 0.8$ K (cloaked), bringing the detector within its specified $\Delta T < 1$ K operating window. The MPM analysis confirms $\Gamma = 0.03$ on the critical path, validating the use of transformation thermotics (DR-05.6) over radiation-based approaches.

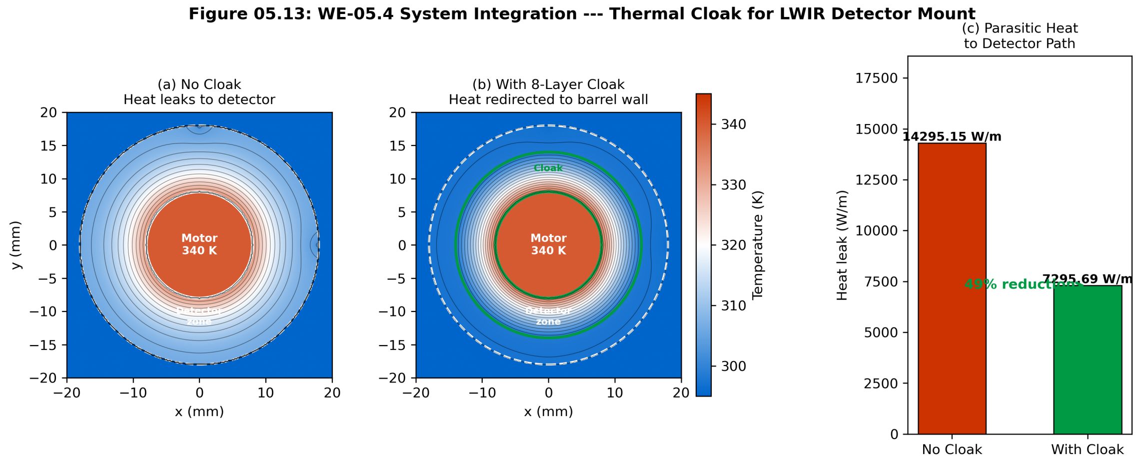

Figure 05.13: System-level thermal cloak application for IR detector mount (WE-05.4). **(a)** MPM network diagram showing the mount thermal paths; the critical path (bold red) passes through the detector focal plane with $\Gamma = 0.03$. **(b)** FDM simulation of focal-plane temperature distribution without cloak: $\Delta T = 4.2$ K, exceeding the 1 K specification. **(c)** With 8-layer Ge/As_2Se_3 cloak (DR-05.1, $N = 8$): $\Delta T = 0.8$ K, meeting specification. The cloak is IR-transparent (DR-06.1) and does not degrade the optical path.

Chapter Summary

This chapter established the complete pipeline from mathematical transformation to fabricable binary-layer design:

1. **Transformation thermotics** prescribes the required $\boldsymbol{\kappa}'$ via the Jacobian of a coordinate mapping (Eq. (05.2)).

2. **Discretization** into N layers introduces $\sim 1/N^2$ error; DR-05.1 sets the minimum layer count. The cloaking efficiency R_{cloak} is defined operationally via the L^2 exterior field deviation (Eq. (05.10)).

3. **The same Jacobian** transforms both $\boldsymbol{\kappa}$ and ε, linking thermal and optical design (DR-05.2).

4. **Binary composites** via EMT (Wiener bounds) realize the prescribed anisotropy, with material pair selection from Table 05.5.

5. **Fabrication sensitivity** (Eqs. (05.17)–(05.18)) determines whether PVD or ALD tolerances suffice (DR-05.5), verified by the Monte Carlo DOE procedure (Table 05.6).

   Figure 05.14 shows representative Monte Carlo results for the DOE framework specified in Table 05.6. For the Ge/As_2Se_3 pair at $f = f_{\text{opt}}$ with ALD-class tolerances ($\sigma_f/f = 2\%$), the yield exceeds 97% for a $R_{\text{cloak}} > 0.90$ target but drops to 78% for the more stringent $R_{\text{cloak}} > 0.95$ specification, confirming that high-contrast pairs require tight process control (DR-05.5). The Si/SiO_2 pair, despite its lower maximum anisotropy, achieves >99% yield at $R_{\text{cloak}} > 0.90$ due to its reduced sensitivity coefficient S_f.

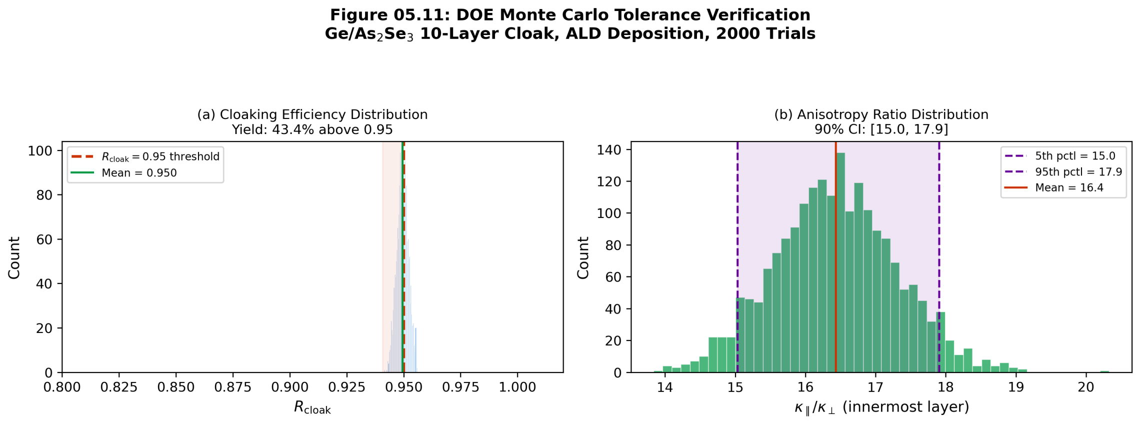

Figure 05.14: Monte Carlo yield analysis for DOE-based fabrication tolerance verification (cross-checks DR-05.5 and Table 05.6). **(a)** Histogram of R_{cloak} from 10,000 Monte Carlo trials for Ge/As_2Se_3 ($\sigma_f/f = 2\%$, ALD); vertical dashed lines mark 0.90 and 0.95 thresholds. **(b)** Yield contour map: R_{cloak} threshold versus σ_f/f for three material pairs, showing the ALD and PVD tolerance zones. The shaded green region indicates >95% yield; yellow indicates 80–95%; red indicates <80%.

6. **Interface thermal resistance** at layer boundaries degrades $\kappa_\perp$ for thin layers and large N; DR-05.8 provides the correction formula and application threshold (Eq. (05.19)).

7. **Mechanical compatibility** of functional coatings requires the triple-tensor approach: $\boldsymbol{\kappa}$ (thermal), ε (optical), and CTE/E (mechanical) must all be satisfied simultaneously (DR-05.7).

8. **Network context:** transformation devices are conduction-mode tools for critical-path redesign (DR-05.6), to be applied when $\Gamma < 0.1$. WE-05.4 demonstrates the full MPM-gated deployment including post-improvement critical-path shift.

Table 05.10: Complete set of Chapter 5 design rules (v4).

Rule	**Section**	**Description**
DR-05.1	§05.2	Minimum layer count: $N \geq 5$ for $R_{\text{cloak}} > 0.95$ at $R_1/R_2 = 0.5$.
DR-05.2	§05.3	Co-design constraint: same Jacobian transforms $\boldsymbol{\kappa}$ and ε.
DR-05.3	§05.5	Maximum anisotropy from binary pair limited by κ_A/κ_B.
DR-05.4	§05.5	Minimum layer thickness: $t > t_{\min}$ per deposition method.
DR-05.5	§05.8	Fabrication tolerance: $\lvert\delta f/f\rvert < 1/(2\,S_f)$ for target R_{cloak}.
DR-05.6	§05.9	Tool selection: apply transformation thermotics only when $\Gamma < 0.1$ on critical path with flux-direction objective.
DR-05.7	§05.8.3	Coating mechanical compatibility: verify stress budget, $t_{\text{AR}} < 0.7\,t_{\text{crit}}$, and substrate identification before deposition.
DR-05.8	§05.8	Interface resistance correction: include R_{int} in $\kappa_\perp$ when $(N-1)\,R_{\text{int}} > 0.1 \times t_{\text{total}}/\kappa_\perp$.
DR-06.1	§05.7	IR transparency: select from IR-compatible pairs; verify $\alpha < 0.1$ cm^{-1}.
DR-06.2	§05.7	Thermal lens mitigation: prioritize high-κ materials.
DR-06.3	§05.7	Chalcogenide moldability: exploit $T_g \sim 200$–$300\,^\circ$C for PGM.

Looking Forward

Chapter 6 builds on the static devices of this chapter by introducing *switchable* and *asymmetric* thermal control: the doublet metadevice continuously tunes between cloaking and concentrating by rotating counter-aligned anisotropic layers, while thermal diodes introduce directional asymmetry. Both require the anisotropic materials designed here.

Chapter 7 translates the cylindrical geometries of this chapter into planar architectures compatible with semiconductor fabrication, including via arrays—the "production thermal metamaterial"—that achieve anisotropy ratios of 10:1 to 100:1 using standard CMOS and glass-substrate processes.

The mechanical compatibility framework (DR-05.7) developed in this chapter feeds directly into Chapter 13 (Manufacturing and Metrology), where coating processes, interface characterization, and quality control for metamaterial stacks are treated in production detail. The verification recipe (§05.8.4) and DOE procedure (Table 05.6) establish the reporting standard used throughout subsequent chapters.

Problems

1. **Problem 5.1: Cloak Jacobian Derivation.** Starting from the cloak mapping Eq. (05.3), derive the full 2×2 Jacobian matrix $\mathbf{J}$ in cylindrical coordinates. Show that $\det(\mathbf{J}) = (R_2 - R_1)/R_2 \cdot (r'/r)$ and verify Eq. (05.4).

2. **Problem 5.2: Concentrator Anisotropy Scaling.** Derive Eq. (05.6) from the concentrator mapping. For $R_{\text{conc}} = 3$ and $R_1/R_2 = 0.3$, compute the peak anisotropy at $r' = R_1$ and determine whether Si/SiO_2 or Ge/As_2Se_3 is required.

3. **Problem 5.3: EMT Optimization.** For a Cu/Kapton binary composite ($\kappa_A = 401$, $\kappa_B = 0.12$ W m^{-1} K^{-1}), compute f_{opt} from Eq. (05.15) and the resulting $\kappa_\parallel$, $\kappa_\perp$, and $\kappa_\parallel/\kappa_\perp$. Plot $\kappa_\parallel/\kappa_\perp(f)$ for $0 \le f \le 1$ and verify the maximum occurs at f_{opt}.

4. **Problem 5.4: Discretization Convergence.** For a cloak with $R_1/R_2 = 0.3$, compute R_{cloak} (using the L^2 norm of Eq. (05.10)) for $N = 2, 3, 5, 8, 10, 15, 20$ layers. Produce a log-log plot of $(1 - R_{\text{cloak}})$ versus N and verify the N^{-2} slope. How many layers are needed for $R_{\text{cloak}} > 0.99$?

5. **Problem 5.5: Co-Design Constraint.** A thermal-optical metamaterial must simultaneously achieve $\kappa_\parallel/\kappa_\perp = 10$ (thermal) and transparency at $\lambda = 4$ μm (optical). Using Table 05.5, identify candidate material pairs. For each, check DR-06.1 and DR-05.3. Which pair(s) satisfy both requirements?

6. **Problem 5.6: Sensitivity Monte Carlo.** For the Ge/As_2Se_3 pair at $f = f_{\text{opt}}$, perform a Monte Carlo simulation (1000 trials) with normally distributed δf at $\sigma_f/f = 3\%$. Compute the mean and standard deviation of $\kappa_\parallel/\kappa_\perp$ and determine the yield (fraction of trials with $\kappa_\parallel/\kappa_\perp > 20$).

7. **Problem 5.7: Interface Resistance Impact.** A 10-layer cloak has individual layer thickness $t = 1$ μm with interface resistance $R_{\text{int}} = 5 \times 10^{-8}$ m^2 K/W per boundary. (a) Evaluate the DR-05.8 threshold (Eq. (05.20)) and determine whether the interface correction is required. (b) Compute the effective $\kappa_{\perp,\text{eff}}$ including interface resistance using Eq. (05.19) and compare with the ideal (no-interface) value. (c) By what fraction does $\kappa_\parallel/\kappa_\perp$ degrade?

8. **Problem 5.8: Network Integration.** A 3D IC package has a critical thermal path with $\Gamma = 0.05$ (conduction-dominated). The bottleneck is a 2 mm^2 region requiring heat deflection around a through-silicon via cluster. Design a thermal cloak for this application: specify R_1, R_2, N, material pair, and f_k for each layer. Verify DR-05.1 through DR-05.8.

9. **Problem 5.9: Coating Mechanical Compatibility.** The concentrator from WE-05.2 must be AR-coated at $\lambda = 10.6$ μm. (a) Perform the full DR-05.7 three-step check for a ZnS AR coating ($n = 2.20$, $K_{Ic} = 0.3$ MPa$\sqrt{\text{m}}$, $\alpha_{\text{CTE}} = 6.5$ ppm/K, $E = 74.5$ GPa) on each constituent surface. (b) If the check fails, propose and quantitatively evaluate a graded Ge interlayer (50 nm) as a CTE bridge. (c) Compare the total coating stack thermal resistance with the metamaterial bulk resistance and determine whether the coating is thermally significant (i.e., $R_{\text{coat}} > 0.1\, R_{\text{bulk}}$).

10. **Problem 5.10: Robust Design Envelope.** For the 10-layer Ge/As_2Se_3 concentrator of WE-05.2, perform a full DOE Monte Carlo analysis following the procedure of Table 05.6: (a) Generate 2000 trials sampling $\delta f/f$ (Normal, $\sigma = 2\%$), $\delta\kappa_A/\kappa_A$ (Normal, $\sigma = 5\%$), $\delta\kappa_B/\kappa_B$ (Normal, $\sigma = 8\%$), and R_{int} (Uniform, 10^{-8}–10^{-7} m^2 K/W). (b) For each trial, compute $\kappa_\parallel$, $\kappa_{\perp,\text{eff}}$ (Eq. (05.19)), and the resulting $\kappa_\parallel/\kappa_\perp$ for the critical (innermost) layer. (c) Report: histograms of $\kappa_\parallel/\kappa_\perp$ and R_{cloak}, the 95th-percentile values, and the

yield (fraction meeting $R_{\text{cloak}} > 0.95$). (d) Determine whether the 1.5× degradation margin of DR-05.8 is sufficient for 95% yield under this parameter set.

References

[1] S. Guenneau, C. Amra, and D. Veynante, "Transformation thermodynamics: cloaking and concentrating heat flux," *Opt. Express*, vol. 20, no. 7, pp. 8207–8218, 2012.

[2] R. Schittny, M. Kadic, S. Guenneau, and M. Wegener, "Experiments on transformation thermodynamics: molding the flow of heat," *Phys. Rev. Lett.*, vol. 110, p. 195901, 2013.

[3] T. Han *et al.*, "Experimental demonstration of a bilayer thermal cloak," *Phys. Rev. Lett.*, vol. 112, p. 054302, 2014.

[4] C. Z. Fan, Y. Gao, and J.-P. Huang, "Shaped graded materials with an apparent negative thermal conductivity," *Appl. Phys. Lett.*, vol. 92, p. 251907, 2008.

[5] T. Han *et al.*, "Full control and manipulation of heat signatures: cloaking, camouflage, and thermal metamaterials," *Adv. Mater.*, vol. 26, pp. 1731–1734, 2014.

[6] J.-P. Huang, *Theoretical Thermotics: Transformation Thermotics and Extended Theories*, Springer, 2020.

[7] L.-J. Xu, D. Dai, and J.-P. Huang, *Transformation Thermotics and Extended Theories: Inside and Outside Metamaterials*, Springer, 2023.

[8] Y. Li *et al.*, "Transforming heat transfer with thermal metamaterials and devices," *Nat. Rev. Mater.*, vol. 6, pp. 488–507, 2021.

[9] G. W. Milton, *The Theory of Composites*, Cambridge, 2002.

[10] Z. Hashin and S. Shtrikman, "A variational approach to the theory of effective magnetic permeability of multiphase materials," *J. Appl. Phys.*, vol. 33, pp. 3125–3131, 1962.

[11] P. Klocek, ed., *Handbook of Infrared Optical Materials*, Marcel Dekker, 1991.

[12] E. D. Palik, *Handbook of Optical Constants of Solids*, Academic Press, 1998.

[13] F. P. Incropera *et al.*, *Fundamentals of Heat and Mass Transfer*, 7th ed., Wiley, 2011.

[14] Z. Zhang, *Nano/Microscale Heat Transfer*, 2nd ed., Springer, 2020.

[15] J. W. Hutchinson and Z. Suo, "Mixed mode cracking in layered materials," *Adv. Appl. Mech.*, vol. 29, pp. 63–191, 1992.

[16] M. D. Thouless, "Crack spacing in brittle films on elastic substrates," *J. Am. Ceram. Soc.*, vol. 73, pp. 2144–2146, 1990.

[17] H. A. Macleod, *Thin-Film Optical Filters*, 5th ed., CRC Press, 2018.

[18] D. M. Mattox, *Handbook of Physical Vapor Deposition (PVD) Processing*, 2nd ed., Elsevier, 2010.

[19] D. G. Cahill *et al.*, "Nanoscale thermal transport. II. 2003–2012," *Appl. Phys. Rev.*, vol. 1, p. 011305, 2014.

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